Project C

Impact, Policy, and System Analysis

IITA

Research to Nourish Africa

Annual Report 2004

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Preface

IITA's research-for-development agenda is divided into six project themes, around which these project annual reports are prepared. These projects themes address different aspects of attaining sustainable increases in productivity of dominant farming systems and utilization practices in the various agroecologies of sub-Saharan Africa (SSA). Research and training activities carried out in the six projects are being implemented together with national program partners in order to increase the well-being of resource-poor people in SSA through higher levels of food production, better income, nutritional status, and reduced drudgery particularly for women.

Project C
Assessing Impact, Formulating Policy Options, and Systems Analysis

New title since December 2004: Supporting Innovation Processes (SINPRO)

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Project overview

IITA is committed to creating new knowledge and technologies that are directly relevant to the needs and livelihoods of rural Africans and widely applied by them. Ultimately, our research products aim to enhance agricultural productivity and food security, raise the incomes of poor farmers, and conserve natural resources. But to achieve this we first need to understand the social, economic, and policy context in which innovations are adopted or rejected, succeed or fail. This project applies the analytical skills of economists, sociologists, information specialists, geographical information systems analysts, and other social scientists to this task.

The project attempts to assess the environment, both socioeconomic and geospatial, in which agriculture operates. Designing policy options and strategies for building market-driven, environmentally sustainable agriculture in Africa is a top priority. Research focuses on ways to commercialize agro-enterprises, improve commodity supply chains, and disseminate market information for the development of competitive agriculture in order to raise rural incomes and ensure efficient food marketing systems for the rapidly growing urban centers of Africa.

Researchers identify bottlenecks in the supply and distribution of seed, fertilizer, and other farm inputs. They examine how institutional arrangements, such as land tenure, marketing arrangements and credit systems, influence agricultural commercialization; evaluate rural and urban food requirements; and study links between agriculture, human health, and diseases such as aflatoxin food contamination, malaria, and HIV/AIDS. To ensure maximum benefits to Africans and efficient use of research funds, the project designs strategies for scaling up the adoption of innovations from individual test villages to entire agroecoregions and beyond. System modeling and geographic information systems are important tools in all these threads of our work.

Within IITA, the project is also responsible for developing and applying methods for setting research priorities and for assessing the impact of the Institute's research, thereby providing essential insights and other feedback to the R4D cycle. Outside IITA, the project works with national agricultural research and extension systems to strengthen their capacity for impact, policy, and systems analysis.

Intermediate project goal

To generate and transfer knowledge for the development of improved food systems, markets, institutions, and infrastructure to accelerate agricultural growth, poverty reduction, improved rural livelihoods and sustainable developments in sub-Saharan Africa (SSA) in partnership with NARES, agricultural research institutes (ARIs), NGOs, the private sector, and other stakeholders.

Project purpose

NARES, NGOs, ARIs, and international scientists effectively work together in collaboration with farmers to define appropriate recommendation domains within and beyond key benchmark areas, facilitate widespread promulgation and adoption of technologies, and assess costs and benefits of research investments.

In partnership with policymakers, NARES, NGOs, and international scientists recommend and support policies that allow farmers to exploit new technologies and opportunities in order to improve their rural livelihoods for the overall development of SSA.

Project outputs

- Output 1: Prioritization of constraints to agricultural systems productivity as well as opportunities and farmers' strategies to improved rural livelihoods at field, farm, and landscape level.
- Output 2: Generation and dissemination of policy and institutional options to improve crop and crop–livestock systems and the functioning of input–output markets.
- Output 3: Development of methodologies and strategies for scaling up and scaling out of best-bet technologies using recommendation domains that are based on key biophysical, socioeconomic, developmental, and cropping systems characteristics.
- Output 4: Documentation of the impact on rural and urban poverty and the environment of research by IITA and its partners.
- Output 5: Strengthened NARES capabilities for maintaining policy, impact, and socioeconomic research.

Project rationale and strategic background

The project goal is to generate and transfer knowledge leading to the development of improved agri-food systems, markets, institutions, and infrastructure for accelerated sustainable development and reduced poverty in sub-Saharan Africa. Efforts are conducted in partnership with national research and extension services, ARIs, NGOs, the private sector, and other stakeholders.

To achieve this goal, the project focuses on five interrelated thrusts:

1. *Prioritization of constraints and opportunities.* Priority setting is achieved through diagnosis and characterization studies of production and farming systems at relevant spatial scales including field, farm, and landscape levels.

2. *Policy studies.* Studies on agricultural policies, agrarian institutions, rural infrastructure, pricing policy, input supply, output markets, and food demand systems are important for guidance in scaling up innovations and developing policy options to overcome market and institutional failures which inhibit the uptake of agricultural innovations.

3. *Scaling-up and extrapolation of research results.* Scaling-up and extrapolation of research results from selected benchmark sites to extrapolation domains (in particular in the forest zones) focuses on the application of geographic information systems (GIS), remote sensing imagery of resource degradation issues, and decision support systems (DSS). This output also contributes to systems analysis at a more regional level.

4. *Impact studies.* Adoption studies, ex-ante impact assessment for priority setting and planning, and ex-post impact assessment measuring changes in social welfare comprise the bulk of the research activities on impact.

5. *Capacity development.* Capacity development is mainly focused on strengthening partners' capacities to undertake socioeconomic research activities.

These thrusts are not mutually exclusive. For example, characterization studies and constraint prioritization provide baseline information for impact assessment as well as helping to set the Institute's research agenda in the biophysical sciences. Similarly, studies focused on policy and the institutional setting, provide important information for evaluating the adoption potential of new technologies and may prove to be serious constraints that require redress to reach development goals.

Attaining the basic CGIAR goals of poverty reduction, food security, and environmental protection, requires good policies, adequate institutions, and infrastructure and the development of suitable technology. The interaction of these factors determines the potential for technology diffusion and adoption, commercialization, diversification, and intensification, all important processes with serious implications for CGIAR goals in Africa. Improved technologies, when developed specifically at the demand of farmers, are a key component of development strategies for poverty eradication.

In an increasingly liberal global economy, market and food consumption studies are important for shaping research priorities and for developing strategies to increase the commercialization, diversification, and intensification of African agriculture.

While appropriate policies can influence incentives for the adoption of new agricultural technology, they are not sufficient to ensure poverty eradication and food security. Estimates of the supply response of crops to price and non-price incentives in sub-Saharan Africa have shown that supply elasticity—the reaction of production of agricultural products on exogenous factors—is higher for investments in rural infrastructure than for market prices. The reason for this is that an improved infrastructure reduces marketing costs and therefore

increases revenues at the farm gate (or farm gate prices). This suggests the importance of improving institutional and infrastructural investment in achieving rural development. Most farmers in sub-Saharan Africa still struggle with inadequate seed multiplication systems, non-functioning or inefficient input and output marketing systems, weak extension systems, thin credit markets, poor educational facilities for their children, a lack of health services, and inappropriate property rights. All of these non-price factors negatively affect the investment capacity in the various types of capital critical for agricultural development and the uptake of new agricultural technologies.

Understanding the relationship between policies, institutions, and infrastructure, and agricultural innovation at various levels (field, household, community, national, and regional) is the main focus of strategic research within the project. The project cuts across various agroecological zones and links with all other projects within the institute. Outputs from the project are fed into other projects for strategic planning as well as technology development and stimulating a policy environment that is conducive to rapid and widespread adoption.

2004 changes in the project

At the end of 2004, the title of project C was changed to "Supporting Innovation Processes" (SINPRO). The outputs have been re-organized to the five outputs as reflected above, namely:

Output 1: Innovations to enhance the profitability, equity, and efficiency of production and marketing systems in the agricultural sector

Output 2: Recommendations for policies, institutions, and legal frameworks supportive of generated innovations and the innovation process

Output 3: Methodologies for scaling up, and scaling out of IITA research innovations

Output 4: Research priority setting and impact assessment of innovations

Output 5: NARES capacity building for social science and spatial analysis

Report according to outputs

Output 1. Innovations to enhance the profitability, equity, and efficiency of production and marketing systems in the agricultural sector

Activities completed in 2004

Resource optimization strategies under differential risk attitudes among maize farmers in Kaduna State, Nigeria

By Olarinde, L.O. (PhD Student, UI), V.M. Manyong, and J.O. Akintola (UI)

Introduction

In the last three decades, many agricultural programs were put in place through agricultural policy instruments to improve the productive capacities of agricultural resources in Nigeria. These were to increase the production of food crops including maize. Despite these efforts, records show that maize yields in farmers' fields have remained consistently lower than anticipated. Some of the reasons adduced for the low yields include: little proportion of area sown to hybrid maize and low rates of adoption of recommended practices. Generally, these problems have been attributed to the absence of a prior analysis of farmers' attitudes towards risk inherent in new technologies. This study therefore examined the attitudes of maize farmers towards risks as they affect their optimal resource allocation.

Methodology

The study was conducted in Kaduna State, Nigeria. A two stage sampling technique was used to select 400 respondents from the four Agricultural Development Project zones. The analytical tools employed include descriptive statistics, ordinary least squares regression (to determine the risk parameter k, for each respondent), discriminant analysis and linear programming models.

Findings

The analysis showed that 7.8% of the sampled farmers were risk-preferers, 41.7% risk-neutral, and 50.5% risk averters. Of the 18 socioeconomic, institutional, and farm specific variables hypothesized to be responsible for the above categorization, 13 (about 72%) were significant at $p < 0.05$. Results of the linear programming models revealed that farmers' observed risk attitudes matched the theory that they allocate resources to different crop farm plans by striking balances among competing objectives of (a) increasing returns, (b) satisfying family subsistence and (c) generating farm income equal to or higher than farmers' income thresholds. Maximum gross returns realizable from the most optimal, alternative, and efficient crop farm plans were ₦42,571.01/ha for risk-preferers, ₦49,945.18/ha for risk-neutral, and ₦19,021.79/ha for risk-averters (farmers). The corresponding shadow prices or marginal value products on the fully used resources were: ₦1.00 of off-farm activities for the risk preferers, ₦0.60 of own capital, ₦0.60 of borrowed capital, and ₦1.00 of off-farm activities for the risk-neutral farmers, while shadow prices of ₦17, 648.78 of land resource and ₦1.00 of off-farm activities were recorded for the risk averters. These values indicate by how much the farmers would have increased their gross returns if they had decided to invest additional Naira on each unit of the fully used resources. The linear programming analysis further showed that with only the profit maximizing objectives, farmers' returns are high, but with additional objectives of meeting family subsistence needs

and generating income not lower than the farmers' threshold income, the farm returns decrease but they are more efficient.

From the study, alternative and efficient crop farm plans that are normatively sustainable and consistent with the risk attitudes of the three categories of the farmers in the study area are recommended. These alternative farm plans provide better resource optimization strategies and they have implications for minimizing input costs and maximizing farm profit. It is also recommended that appropriate policy mix and technological innovations, e.g., some tax holidays and conducive interest rates that enhance the acquisition of the needed basic farm inputs be put in place. These incentives would render farmers' risk preference flexible and make them operate within the normative, alternative, and efficient farm plans. This is possible through the combined efforts of research institutions, government and extension agencies, and the private sector.

Socioeconomic determinants of aflatoxin risk and agricultural technologies as possible solutions

By Kerstin Hell, Ousmane Coulibaly in collaboration with K. Djade

Introduction

Maize (*Zea mays* L) plays a key role in the food security of the populations of Bénin and Togo. It is very sensitive to the *Aspergillus* mushroom which develops to it easily by producing mycotoxins in particular aflatoxin which is very dangerous for human and animal health.

Methodology

A survey was led in coastal savanna and Southern Guinean Savanna of Togo and Bénin in order to determine the socioeconomic factors which influence the risks of contamination by aflatoxin through farmer's perceptions and attitudes on the moulds and aflatoxins. Two types of structured questionnaire (man and woman) were managed with a sample of 132 households chosen randomly in the 6 villages chosen in the two agroecological zones.

Findings

The study revealed that more than 90% of respondents recognize corn of bad quality through signs such as the presence of mildewed and/or rotted or faded grains, the presence of insects or larvae, and the presence of farinaceous dust due to the insects. The socioeconomic factors being able to influence the risks of contamination by aflatoxin are inter alia: the too strong frequency of corn consumption and the non diversification of the food, lack of quality control of stored corn, lack of information and sensitizing. The econometric Logit model was used to analyze the determinants of the adoption of the use of insecticides, more particularly of Sofagrain (Deltaméthrine 1.5%+ Pyrimiphos 0.5 %) in maize storage. The variables such as the variety of cultivated corn, the membership of a country grouping, the agroecological zone, the maize surface cultivated, and the mode of storage of corn determine to a significant degree the adoption of the use of insecticide in maize storage.

Farmer-to-farmer diffusion of improved cowpea varieties and the production efficiency of adopters in northern Nigeria

By Arega D. Alene, V.M. Manyong, A. Tipilda, B.B. Singh

Introduction

The International Institute of Tropical Agriculture (IITA), in collaboration with the Institute of Agricultural Research (IAR) and the Kano State Agricultural and Rural Development Authority (KNARDA), launched a farmer-to-farmer improved cowpea seed diffusion project in 1997. In an effort to exploit the yield potential of improved cowpea varieties, on-farm demonstration has aimed to promote an improved cowpea package of improved variety seed, insecticide, inorganic fertilizer, and a cropping pattern involving two rows of cereals by four rows of improved cowpea. It is believed that the number of farmers using improved cowpea has considerably increased over the years between 1997 and 2003. While a growing number of beneficiary farmers over the years certainly shows the effective dissemination of improved cowpea seed, little is known about the dissemination of the package as a whole and the resulting yield and efficiency variation among adopters. The objective of this study was thus to investigate the extent and determinants of production efficiency of adopters of improved cowpea in Kano State, northern Nigeria.

Methodology

The study used data collected from 253 farmers in Kano State who have adopted improved cowpea varieties. A translog stochastic frontier and inefficiency model was used to investigate the extent and determinants of production efficiency of adopters.

Findings

The average production efficiency of adopters of improved cowpea was estimated to be 63%, implying that adopters could increase cowpea production by an average 37% if they used the "best" management practices. The results revealed important efficiency differences between the lead farmers (74%) who have contacts with researchers and the follower farmers (61%) who get technology and information from the lead farmers. Differential adoption of the package provides much of the explanation for yield variation among adopters. The component largely missing, and hence accounting for much of the inefficiency, was the crop management technology relating to the cereal–cowpea cropping pattern. No efficiency variation was attributed to the source of technology and information such as whether improved cowpea was obtained from researchers or lead farmers. This suggests that farmers could produce and disseminate improved seeds as effectively as the formal system, but may not be as effective in disseminating crop management information. Technology source has rather had an indirect influence through its effect on package adoption. Innovative ways of disseminating crop management technological information should thus be integrated into the farmer-to-farmer technology diffusion in order to better exploit the yield potentials of improved cowpea varieties in northern Nigeria.

Baseline socioeconomic survey of PROSAB project area

By P.S. Amaza, A. Adejobi, Y. Bila, and A.C. Iheanacho

Introduction

The goals of PROSAB project are to contribute to improved food security, reduce environmental degradation, improved sustainable agricultural production through the transfer of improved agricultural technologies and management practices to male as well as female farmers, improved market access, developing a more enabling policy environment, and enhancing the capacity of its partners. As a result, a baseline survey was carried out in the project area to provide sex-disaggregated baseline data on socioeconomics, resource-use patterns, and market opportunities and their effects on land degradation and agricultural productivity in potential targeted project communities.

The major baseline indicators or criteria for measuring and monitoring farmers' economic status and progress include:

- Production characteristics such as the gross margin from households' crop production and livestock
- Livelihood diversification indices such as households' engagements in non-farming activities and enterprises
- Households' access to improved farm inputs
- Food expenditure and consumption by farm households
- Farm households' food security and poverty status
- Differentials in access to resource according to gender.

Socioeconomic measurements as well as food security and poverty computations were provided to help support and explain the various baseline indicators. These will provide information on the capabilities and constraints under which the farmers, their spouses, and family operate to achieve the desired goal of improved household welfare.

The objectives of the baseline survey are to:

- examine the socioeconomic characteristics of households in the project area including major farm and non-farm economic activities.
- examine crop and livestock production characteristics of the households.
- determine the type and level of inputs used in crop and livestock production.
- estimate farm and non-farm income of the households.
- examine the food expenditure and consumption pattern of the households.
- determine the food security status of the households.
- analyze the factors that influence food security in the project area.
- estimate the poverty measures of households in the project area.
- examine the intensity of poverty in the project area.
- examine the determinants of poverty in the project area.

Methodology

Survey design

The baseline field survey was carried out in the 30 communities covered by the PROSAB project in Borno State. Borno State, located in northeast Nigeria covers an area of 69435 km^2. The State is demarcated into four agroecological zones: southern and northern Guinea Savanna in the south, Sudan Savanna in the south and central, and the Sahel in the north. The annual precipitation ranges from less than 600 mm in the north to 1500 mm in the south. Rainfall varies from year to year but has tended to decrease over the last two decades. The population of Borno State is estimated at 3.35 million people[1]. More than one million people live in Maiduguri, the State capital. The population of the project area is estimated to be about 1.67 million.

The project area is located between Latitude 10° and 12° north of the Equator and Longitude 11°.30 and 14° east. It consists of *Biu, Damboa, Hawul* and *Kwaya Kusar* local government areas (LGAs). Numerous ethnic groups and cultures characterize it with approximately 80% of the population being small-scale farm producers. Agriculture is the main source of livelihoods.

The baseline study area is characterized by a relatively humid weather as compared to the northern part of the state hence it permits for good agricultural activities. The annual rainfall ranges from 700 to 1200 mm with about 140 to 150 days of intensive rainfall. The raining season starts in April and lasts till October. The humidity is about 49% with 2034mm evaporation rate per annum. The temperature varies from 15° to 20°c during the harmattan period (i.e., December to February) and also from 32 to 38°c during the hot dry season (i.e., March to May).

The agricultural activities can be categorized into crop production, animal husbandry, and agricultural marketing. The cropping pattern in the project area is almost uniform probably due to the similarities in vegetation of the area. There are two cropping seasons, one which starts with the onset of the heavy rains and then the dry season cropping starts soon after harvesting of the rain-fed crops. Cropping may be sole, multiple, mixed, and relay cropping. The crops may be grown in rotations depending on preference. Major crops grown in the area include: maize, sorghum, cowpea, groundnuts, and vegetables. The animals reared include cattle, sheep, goats, pigs, and poultry.

Sampling technique

The baseline data were obtained through a household survey conducted between June and August 2004. The main instruments for data collection were well-structured questionnaires administered on households by trained enumerators under the supervision of the PROSAB Agricultural Economist and collaborators from the Department of Agricultural Economics at the University of Maiduguri.

Four (4) LGAs were covered for the purpose of data collection. The data were collected from 39 communities spread across the 4 LGAs in the project area. First, the 30 communities that were identified and selected during the livelihood analysis for the implementation of project activities were purposively included in the sampled communities. In addition, 9 communities, where it is expected that the project activities will scale-out were also

[1] 2001 estimation.

selected for the data collection exercise. In each selected community, a random selection of households was carried out using sampling proportion to size procedure (40 respondents in small size communities; 80 in medium size communities, and 120 in large-sized communities). To ensure gender balance, the survey was designed such that 25% of the respondents were female. This implied that 25% of the interviews were done with female household members and not specifically the household head as in the 'general' survey. The survey instrument was also modified in some areas to incorporate some gender specific issues. A total of 2250 questionnaires were administered out of which 2205 questionnaires were retrieved and used in the descriptive statistics. However, 1998 questionnaires were used in food security and poverty analysis due to incomplete information on some of the dropped questionnaires.

Data analysis

A combination of analytical tools was employed in the analysis of data. These included descriptive statistics (means, frequencies etc.), budgeting, cost-of-calorie calculations, Logit and Tobit regression models.

Descriptive statistics was used to examine the households' socioeconomic characteristics, crop, and livestock production patterns, etc. The budgetary technique was used to determine the profitability of food crop enterprises. It provided actual information on farm-input use and costs, output and prices, and farmers' gross margins.

To measure the food security status of households a cost of calorie index was constructed. The cost-of-calories (COC) method proposed by Greer and Thorbecke (1984) was used in this study to determine the food security line. The method yields a value that is usually close to the minimum calorie requirements for human survival. The Logit model was used for the determinants of household food insecurity.

To determine the poverty status of the households in the study area, a poverty line was constructed using the two-third mean per adult equivalent expenditure, below which households are classified as poor or non poor. For the determinants of household poverty, a Tobit regression analysis was conceptualized and the full model, which was developed by Tobin (1958), following McDonald and Moffit (1980), and as adapted by Omonona (2001) was applied in the poverty analysis.

Findings

The major findings from the analysis of the baseline survey are as follows:

It was revealed that the average sampled head of household was a male, about 46 years of age and more than 70% of the population was in the active and productive age. The average number of years of education of head of household was 5 years, a Moslem, usually married and was polygamous (having more than 2 wives). An average household head was basically into farming (though the highest income may not be from farming) and had a fairly large household size of between 6 and 11 adult male equivalent. An average household cultivated about 3 hectares of land in at least 2 or more different fields.

The mean monthly per adult equivalent household expenditure was ₦3670.00 and the poverty line was ₦2446.67. This poverty line revealed that about 67% of the households in the project area were poor (Table 1).

Table 1. Distribution of households by poverty status.

Category	Frequency	Percentage
Poor	1332	66.67
Non-poor	666	33.33
Total	1998	100.00

Source: Survey

Table 2. Summary statistics and food insecurity measures among households in the project area.

Variable	Value
Cost-of-calories equation	Constant = 4.154 (0.534)
	Slope coefficient = 0.0019 (0.0004)
FAO recommended daily energy levels (L)	2250 Kcal
Food insecurity line Z: cost of the minimum energy requirements per adult equivalent	₦ 63.71 per day
	₦1975.01 per month
	₦ 23700.12 per year
Head count (H)	0.58
Aggregate income gap (G)	-375.74

Source: Calculations from OLS estimates and cost-of-calories equation.
Figures in parenthesis are t-values.

The most important livelihood strategies pursued by the households identified in the study area include agricultural production (mainly crops) and small and micro household enterprises.

The higher the household size and child dependency ratio of the households were, the higher was the probability of food insecurity as well as the intensity of poverty among rural households in the project area. A summary statistics of food insecurity measures among the sampled households is presented in Table 2.

Baseline market survey to establish the structure, conduct, and performance of key commodity markets in the PROSAB project areas

By P.S. Amaza, S.K. Msheliza and P.V. Kwaghe

Introduction

One of the key factors to successful and sustained adoption of yield enhancing agricultural technologies such as the ones transferred by PROSAB project is an assured market outlet. Marketing and production are interdependent. Producers must be convinced that a remunerative market exists for their products, particularly 'new' products, before they can be induced to produce commercially. An attractive market prospect combines a 'good' price with and an assured sales outlets. This simple, self-evident truth has sometimes been overlooked in agricultural project planning. Marketing projects have failed both because an adequate price incentive was not backed up by a necessary addition to the market

infrastructure (processing, storage, and transport facilities) and because infrastructure was provided without a sufficiently attractive price.

In addition to attractive market price and assured sales outlets for their crops, farmers' involvement in adoption and commercial production could be enhanced by the timely availability of appropriate inputs at reasonable prices. The success in adoption of production enhancing technology is highly dependent on a well-developed and efficient input and output marketing strategy.

One of the objectives of PROSAB is to increase agricultural productivity and incomes of male and female farmers. This prompted the survey of the existing market conditions; identification of input and output dealers and processors within and outside the project area with the aim of developing an appropriate agricultural input and output marketing strategy for males as well as female farmers in the project area.

The objectives of the study are:

i. To undertake a rapid appraisal survey to establish the structure, conduct and performance of the key commodity markets, including livestock in the project area and specific niches occupied by women.

ii. Carry out a rapid appraisal in selected villages to obtain additional information on existing marketing and transportation arrangements; mechanism for commodity price establishment; regional trading pattern and processing potential; and engendered barriers to entering the market.

iii. Identify within and outside the project area male and female agricultural input dealers; commodity dealers and processors; assess their requirements in terms of product quality/quantity; delivery schedules and pricing and undertake a strength and weakness analysis.

Methodology

Survey design

The study area consists of agricultural input and output markets within and outside the PROSAB project area. Markets and communities visited for the survey within the project area include: Biu, Miringa, Sobon Gari, Kwaya Kusar, Shaffa, Marama, Sabon Kasuwa, Kukurupu, Tasha Alade, Mbulatawiwi, Damboa, and Mandaragirau. While markets outside the project area surveyed include: Gombe, Maiduguri, Damaturu, Potiskum, Jos and Kano. These are the immediate major towns which have direct links with the project area in term of marketing of agricultural inputs and outputs. In these cities are large-scale dealers of various agricultural inputs and outputs as well as industrial processors of agricultural commodities who purchase outputs or supply inputs in the project area.

Data collection

A team of eight (8) multidisciplinary researchers from CRED (NGO collaborator) collected data over a period of 50 days (September–October 2004). First, markets within the projects areas were visited where data on the market structure, conduct, performance, transport arrangement and infrastructural facilities available were obtained. Also obtained from these market communities were data on input and output dealers as well as agricultural commodity processors. Similarly, markets outside the project area were also visited to obtain data on agricultural commodity requirements of major industrial processors and

large-scale output dealers. Also obtained were information from major suppliers of agricultural inputs such as fertilizer, agro-chemicals, storage facilities, veterinary drugs, and animal feeds. The choice of markets outside the project area was informed by the major trading partners of crops and livestock produced in the project area.

One hundred and fifty two (152) agricultural output processors were contacted; 140 of them are micro-processors, which are meanly located within the project area, 12 are industrial processors outside the project area.

Two hundred and two (202) agricultural commodity dealers that trade in commodities such as maize, cowpea, soybean, sorghum, groundnuts, cattle, goats, and sheep were interviewed. One hundred and twenty one or 60% of sample operate in the project area, while the rest are outside i.e., in Gombe, Maiduguri, and Damaturu. Purposive sampling was adopted to include male and female commodity dealers. A total of 95% of the commodity are male dealers and only 5% are female. Commodity dealers were classified according to size based on quantities traded annually.

Seventy two (72) agricultural input dealers within and outside the project area were also contacted and interviewed. Agricultural input dealers included in the research were merchants dealing in fertilizer, agro-chemicals (such as herbicides, pesticides, storage chemicals, and seed dressings), veterinary drugs, livestock feeds, storage facilities (such as woven sacks, polythene bags, and drums), farm implements, and tools. Input dealers were classified into various sizes based on turnover, number of employees, and physical visit to shops; 58 or 80% are small scale, 11 or 15% are medium scale, while 4 or 5% are large scale.

Twelve villages/towns all within the project area were visited on their market days in order to assess the market environment. The team interacted with market officials and marketers on issue relating the markets such as availability and conditions infrastructural facilities, market conduct, performance and structure.

A number of methods were used to obtain data for the study. Primary data was collected using Rapid Results Techniques of focused group discussion, in-depth interviews, and observation. A structured questionnaire was also administered to obtain information from processors, input and commodity dealers, and officials in charge of markets locations. Secondary data was obtained from review of literature and records of market practitioners, NGOs, CBOs, and government agencies (see e.g. IFDC, 2001).

Data analysis

Data obtained were analyzed using descriptive statistics such as frequencies, percentages, and cross tabulation. Conclusions were drawn from the results and recommendations made on strategies to help develop the agricultural marketing system in the project area.

Findings

1 Seventy two (72) agricultural inputs dealers were covered in the survey. They are of various sizes and deal in diverse agricultural inputs: agro-chemicals, fertilizers, seeds, livestock feeds, veterinary drugs, storage/packaging facilities, and farm implements. By way of size analysis; 80% are small scale, 15% medium scale, and only 5% are large scale. 90% are sole proprietorships and partnerships, while 10% are limited liabilities. In terms of sources of supply, 85% obtained their supply locally through other traders, while 10% buy from suppliers in Lagos, Onitsha, and Kaduna. About 5% import from

other countries; 40% of the dealers maintain book accounts and 60% being small-scale traders have no relationships with banks. The main source of information to dealers are through farmers' enquiries. They help dealers to ascertain demand situation. On the other hand traders get information from major suppliers in terms of availability of products, new technology, and prices.

2 Two hundred and two commodity dealers (202) were surveyed within and outside the project area. The major commodities traded included maize, rice, sorghum, cowpea, groundnuts, soybean, pepper, and livestock such as cattle, sheep, and goats. The study revealed that within the project area 70% of the dealers are small-scale (dealing in less than 20 tons); 25% are medium-scale: (20–50 tons); and 5% are large-scale (more than 50 tons). In each market the dealers are grouped into associations and cooperatives, mainly to protect the interest of their members when it comes to taxation and demands from government agencies. The associations do not engage in price fixing or other marketing activities. Nevertheless, commodity dealers have better quality information on market conditions than farmers. Large-scale dealers seem to have more access to market information than small-scale dealers.

3 The major agricultural products processed are groundnuts and soybean to obtain oil and cake, rice to obtain per boiled rice, maize is milled into flour for human consumption and livestock feed, and yellow maize is processed into corn flakes, which is a breakfast cereal. The survey shows that within the project area, there are no large-scale (industrial) processors. There are, however, micro enterprises in all the surveyed markets. 140 micro-processors were contacted in the study and all of them are within the project area. Twelve (12) large-scale (industrial) processors were contacted outside the project area in Maiduguri, Potiskum, Jos, and Kano. These include 5 flour mills, 6 vegetable oil and livestock feed processors, and 1 food and confectionary. The information sought from processors included installed capacity, pattern of demand, and buying schedules of agricultural output. The result shows that all (100%) micro-processors, of which over 70% are women obtained their raw materials locally within the project area. The major problems faced are lack of credit and storage facilities. Approximately, 90% of the micro-processors have no bank accounts at all. However, the large-scale industrial processors obtain their raw materials mainly from Plateau and Kaduna States for maize and soybean; Borno, Kano, Katsina, Adamawa and Zamfara States for groundnuts; and Benue State for soybean.

4 Twelve (12) markets were selected and surveyed during their market days. Discussion and interviews were conducted with market officials and selected traders who market agricultural inputs and outputs. Analysis of the market structure of the input and output markets revealed that the number of buyers and sellers are not equal. In the input market, it is observed that the number of buyers who are mostly farmers outnumber the sellers. While in the output markets the number of sellers who are also farmers outnumbered their counterpart buyers, the middlemen. The study revealed that within the project area there is no legal barrier to entry or exit of either of the two markets. Both women and men participate in the marketing processes. The only barrier reported by few market officials interviewed is in the input market that demands huge capital requirement and technical know-how. Concerning market information, all those interviewed reported that the source of market information available to marketers particularly farmers is informal. The method of dissemination of the information source

is mainly through personal contact with friends and association members, and 91.7% of them are not satisfied with how the informal market information is disseminated.

Organizing meetings between the private sector with farmer group representatives to facilitate establishment of agreeable contract arrangements within the PROSAB project

By P.S. Amaza, S.K. Msheliza, and P.V. Kwaghe

Introduction

The focus of the market linkage activity was to link farmers' producer groups with market outlets for the sales of produce. The objective of organizing meetings between commodity processors and farmers' group was to facilitate establishment of agreeable contract arrangements for bulk delivery of products. Because the farmers are organized into groups, they are expected to have economies of scale advantages when they sale in bulk.

The objective of organizing meetings between commodity processors and farmers' group was to facilitate establishment of agreeable contract arrangements for bulk delivery of products.

Methodology

Some processors were invited and held meetings with farmers' group, with PROSAB playing a facilitating role. During the report period, two meetings were held between farmers' groups and the private sector (processors) on agreeable contract arrangements for bulk delivery of commodities (maize and groundnuts). The first meeting was between Yobe flour mills and farmers' groups in Miringa to supply 150 tonnes of maize at a supply price of N36 000.00 per metric tonne.

The second meeting was between P.S. Mandridis, a vegetable oil and livestock feed mill located in Kano and farmers' groups in Damboa, Sabon Gari, and Kwaya Kusar to negotiate bulk supply of groundnuts to the oilseed processor in Kano. At the meeting 18 farmers' groups leaders attended the meeting in Damboa; 22 farmers' group leaders in Kwaya Kusar and 20 leaders in Sabon Gari communities.

Findings

In December 2004, Yobe flour mills met with leaders of farmers group in one of the project community (Mirnga), which resulted to the granting of purchase order to maize producers in Mirnga to supply the flour mill with 150 tonnes of maize at a supply price of N36 000.00 per metric tonne.

The farmers' group later decline the offer because the price was rather low when transaction costs were taken into account. Moreover, at the time of the offer, a metric tonne of maize was being sold in the community at N35 000.00 per metric tonne. Thus, when transportation and other transaction costs are taken into account, the farmers will lose. The prices of grains have generally been high in all the project communities due to the effect of drought in some parts of the state. As a result there has been an upsurge in the demand for maize, a major staple in the state, resulting in increasing market prices. In fact, the market price of maize has more than doubled between December 2004 and March 2005.

Survey on banana-coffee systems, part: Economics of farming and marketing systems

by Steffen Abele, Suleiman Okech, Cliff Gold, Caroline Nankinga—in collaboration with A. Nabuye (Coffee Research Institute (CORI, Uganda) A. Karugaba, Mbarara District Agricultural Extension Department, Uganda) and W. Tushemereirwe (Kawanda Agricul tural Research Station, Uganda)

Introduction

The purpose of the study was to provide first insights into the yet under-researched farming systems in Western Uganda that are based on banana and coffee. These two crops are the economic bases for most of the people living in the rural areas of Western Uganda. They provide both cash (coffee and banana) and, in the case of banana, also food and by-products. Yet, the system faces decline: both banana and coffee are affected by pests and diseases that threaten the whole system and the livelihood of the rural communities.

Methodology

The methodology used to obtain the information was the conducting of group interviews on three sites in Mbarara during April/May 2003. Group interviews were held during meetings with 15 to 40 farmers. Further, single farmers were visited on their fields and interviewed in a non standardized way.

Findings

Farm sizes are rather small in all the sites, usually averaging about 0.8 ha (2 acres), but ranging from 0.1 to 47 ha and from 0.1 to 1.6 ha across sites. While in the first area, land is mainly consolidated, in other sites, land holdings are dispersed due to the heritage system.

As far as labor organization is concerned, hired labor is used on all sites, mainly for weeding and pruning, but also for soil preparation. On one site, labor is scarce during the high cropping season, in Ndaija, mainly during the coffee-picking season, while on other sites, labor scarcity does not seem to be much of a problem, probably due to the remoteness that makes off-farm alternatives scarce. Labor is mostly hired for work on cash crops (banana and coffee); for food crops, only for soil preparation.

Cropping patterns are quite similar across sites. The most prevalent system is intercropping, with a few exceptions in banana and coffee that are discussed further below. The main reason for intercropping is land scarcity, and it appears that land-use management in terms of complementary intercrops (e.g., legumes to fix nitrogen) is hardly known amongst farmers; at least it was never explicitly mentioned during the survey. Sole cropping mainly occurs in banana and in coffee systems. Although they were predominantly intercropped in the first two sites, coffee was mainly planted sole in the third site.

Crops differ only slightly from site to site. Apart from bananas and coffee, millet, maize, groundnuts, beans, but also horticultural products such as avocados, oranges, pineapples, mangos, passion fruit, and guavas are grown. In all the sites, one has to discriminate cash and food crops. Coffee and banana are major cash crops on all the sites. Food crops are bananas, cassava, millet, and the horticultural products, whereas cash crops are (according to the farmers' ranking) bananas, beans, coffee, livestock products, especially milk, millet, and peanuts. As stated above, most of these crops are intercropped with at least one other crop. The major part of banana types grown is at all sites of the *matooke* (cooking banana)

type, followed by brewing banana (*mbiri*), large desert bananas (*bogoya*) and, to a smaller scale, small desert or apple bananas (*ndizi*).

As for coffee, the dominant variety is *Robusta*, the higher quality yielding *Arabica* is rarely found. An improved variety of *Robusta*, *Colono*, is also found on the sites, yet to a lesser extent than *Robusta*, as *Colono* requires high management (which is not available due to liquidity constraints and the consequent labor shortage), and is susceptible to all kinds of agro-climatic risks, especially to draught. The biggest problem, apart from management, seems to be coffee wilt. One of the agronomic advantages of coffee is that it can be grown on marginal soils.

Banana and coffee are, beside beans, the two most important cash crops in the region. But this is not the only feature they have in common. These crops interact in many ways, be it in economic terms or in agronomic terms as intercrops.

Concerning the agronomic interaction, we have to first look at the banana–coffee intercropping systems. First, there seems to be a certain competition between the crops on intercropped stands. They obviously compete for soil fertility, while coffee is often believed to deplete soils and eventually kill intercropped bananas. Banana seems to throw shade on coffee, which is advantageous for the younger plants but disadvantageous for older and higher coffee plants. Bananas are also seen to retain moisture and thus provide a certain shelter for coffee in times of draught. Further, both banana and coffee provide mulch for each other; bananas in terms of leaves and cut stems, coffee in terms of coffee-husks (which cannot be spread on coffee due to the consequent spreading of coffee wilt).

The most important economic relationship between the two crops is that they provide seasonal liquidity (especially coffee) for their mutual benefit on management. During the whole year, banana sales provide liquidity for hiring labor for banana and coffee management, coffee provides liquidity for banana management during the coffee harvest season. The same controversy as for agronomic interactions is found when it comes to the discussion of economic features of the two crops and their profitability.

Comparing the mere cash revenues of banana and coffee, coffee generally seems to be the slightly more profitable crop. Here, it can be seen that coffee yields, according to site between 600 000 and 1 400 000 Ush per acre and year in cash, while banana yields between 400 000 and 1 000 000 in cash on the same area. But if the additional amount of food is calculated and the value that banana gives, the picture changes rapidly. Assuming that 15 to 50% of the harvest is home-consumed, we can add a value of about 150 000 to 320 000 Ush to the economic yields of banana, which makes banana about as profitable as coffee, although slightly less. It can thus be assumed that farmers renounce possible coffee profits for the sake of liquidity and food security, although this renunciation is at current prices rather small.

In general, the farming systems on the survey sites can be considered as both market and subsistence oriented. Apart from coffee, of which the marketed share is naturally 100%, be it fresh or dried, for other crops such beans, 50% of the harvest is marketed. In the case of banana, 50 to 85 % of the harvest is sold, depending on the site and, of course, on the relationship between land endowment, land productivity, and the nutrition requirements of the owner and his or her family.

Generally, products are sold at the farm gate, without having a market place. Brokers go around the farms and negotiate the price on behalf of the traders. The problems the

farmers encounter in this system are manifold. Above all, there is a huge asymmetry in information, as banana prices in Kampala are only known by the brokers or traders, not by the farmers. The farmers have to take what they are offered by the brokers, the latter having at least a temporal monopoly at the farm gate. The problems encountered in coffee marketing are mainly similar concerning the weak marketing position of the farmers.

Bananas, especially *matooke*, prices show a significant seasonality. While during most of the year, from September to March, prices range from 2000 Ush to 3000 Ush per bunch (depending on size and quality of the bunch), they go down to 50 to 250 Ush per bunch, during the season when large quantities are harvested, i.e., April to August. Yet, price fluctuations slightly differ across sites. Moreover, additional price deductions up to 25% during the rainy seasons according to poor accessibility through roads are reported.

As far as organizations and marketing boards that could counterbalance these negative developments especially in coffee prices, but also increase the market power in other products' trade are concerned, at none of the three sites such an organization, e.g., a cooperative, exists.

As a conclusion, it can be said that systems, both farming and marketing, are complex and diverse and a lot of research is required to provide optimal support for the development of these systems. It seems that both the economic and agronomic potential especially of intercropping systems is not yet fully exploited. Banana and coffee play a major economic role as cash crops (both) and food crop (banana) on all the sites. Whether they are grown intercropped with each other or cultivated on pure stands, banana and coffee have many economic and agronomic relationships, both positive and negative. Behavior patterns of the farmers are not only profit orientation, but liquidity and food security play an important role in economic decision making. The major constraints to production are land scarcity, poor marketing options and therefore a lack of liquidity, which leads to poor crop management and poor exploitation of the economic potential.

Banana systems' agronomics and economics in Uganda

by Steffen Abele, Piet van Asten, Cliff Gold, and Suleiman Okech

Introduction

Cooking banana (*Musa spp.*) is the most important starchy staple food in Uganda. In 2001, Ugandan per capita consumption of bananas amounted to 188 kg per year. It therefore ranges before crops like cassava, sweetpotatoes, and maize with 107, 89, and 31 kg per capita consumption respectively. However, although overall production has increased in the decade from 1991 to 2001 from 8 million tons to almost 10 million tons per year, per capita consumption has decreased from a mid-nineties level of more than 200 kg per capita. Most of the bananas are consumed in rural areas, probably from own subsistence production. Market effective demand is observed in urban areas such as Kampala and other major towns. It is characterized by low income elasticity and a relatively low price elasticity which is typical for starchy staples. Alongside with this, real market prices have declined steadily in the respective decade.

In addition to from the above described phenomena, producers claim to suffer from low farm gate prices due to low negotiation power with traders and their agents, but also due to difficult postharvest handling, high perishability, and poor infrastructure, all of this resulting in high postharvest risks. Prices shift strongly across annual seasons, with

peaks towards the festive seasons at the end of the year and lows during the high banana production season in mid-year.

Production problems include declining soil fertility, pests and diseases, and more recently rainfall shortages in the main production area, Southwestern Uganda.

Methodology

In order to target these problems, statistically sound analysis is needed to determine the effects of exogenous factors affecting banana production and marketing. The study is based on farm trial data from the major banana production zone in Mbarara, Western Uganda. Over 2000 observations from more than 100 trial plots were analyzed using bi-variate descriptive and multiple regression analysis. Objectives were to determine seasonal price fluctuations, factors affecting farm gate prices such as bunch weight, and biotic and abiotic factors contributing to banana yield.

Findings

Analysis shows that seasonal variation has a high effect on farm gate prices. It can be shown that highest prices can be obtained in December, whereas lowest prices are observed between February and July. Harvested quantities show an inverse relation: Most of the harvests occur in the low price season, which is probably contributing to the low prices, while in seasons of high demand such as the festive seasons at the end of the year, the amount harvested drops.

Another important factor affecting the farm gate price per kg of banana is the bunch weight. It can be shown that the larger the bunch, the lower the respective price/kg of the bananas. This contradicts the hypothesis that larger bunches should yield higher kg prices due to easier handling and economies of scale in transport. However, the opposite is the case. This reveals that there might be institutional frictions in buying the bunches at the farm gate, such as a lack of negotiating power by the farmers and the purchase of whole plot harvests without taking into account the individual bunch weights. Another explanation is the difficult transport from the farm gate to the market place or collection centers, which is often effected by bicycle and thus subject to diseconomies of scale. In this context, the optimal bunch weight in terms of prices/kg is around 22 kg.

In the field of agronomics, factors and their influence on banana yields were tested. The factors hypothesized to have an influence on yields were rainfall during different stages of the plant and fruit development, NPK fertilization, as well as the soil nutrient that had been identified as probably the most significant in the descriptive analysis, K.

Results show a significant influence of three rainfall periods, namely within 6–12 months before harvest, as well as in the period immediately before harvest. The K content in the soil seems to have a high impact, much higher than the impact of combined NPK fertilizer. This leads to the conclusion that profitability is higher if only one fertilizer component, K, is applied. The high and significant impact of rainfall suggests that during droughts (e.g., the 2004 drought), irrigation or other water management measures could be used to improve yields. Such measures could also be used to time harvest better to seasons where there are higher prices.

Uganda's banana markets and production systems are subject to a number of constraints. Production has to be kept up to avoid food shortage and price increases. At the same time, any production increase that is not demand driven will lead to a deterioration of prices.

To stabilize and balance the system, measures on both production and marketing are required. The study shows up some intervention points for this. Market analysis shows that high price seasons should be targeted through increased production, e.g., through specific de-suckering methods, while the fact that there is an optimum bunch weight in terms of prices/kg gives information for breeders to target respective weights. Yield analysis shows that through irrigation or water management, yields can be increased, and harvest seasons can be targeted. Fertilization seems to be most profitable if K is applied, and less profitable with a full NPK application.

Ongoing activities

Market opportunities identification study in Southern Sudan

By Steffen Abele

Introduction

The study follows up the first cut study on market opportunities done in 2003. It covers primary production and marketing, as well as processing and trade schemes.

Methodology

Farmers, traders and processors were interviewed in both the crop and the livestock sector, covering all regions of Southern Sudan. Data obtained from the interviews were subjected to a descriptive analysis, mostly bi-variate.

Findings

First results show that despite the decade long distress, farmers in Southern Sudan are very market oriented and cash crops are in line with market demand especially from the domestic market and from Uganda. However, agriculture has to be intensified to meet the growing demand, as it is expected that the Southern Sudanese population will be growing by almost

50% due to the re-migration of internally displaced people into Southern Sudan. Constraints at production level are pests and diseases, as well as lack of (mostly financial) resources. Constraints at the marketing level are poor transport infrastructure, (illegal) taxation, the lack of trading centres, and seasonal over production that leads to low prices.

Output 2. Recommendations for policies, institutions, and legal frameworks supportive of generated innovations and the innovation process

Activities completed in 2004

Improvement of banana marketing and utilization in Uganda

By Steffen Abele

Collaborators: National Agricultural Research Organization, Drs Tushemereiwe and Kalyebara, Makerere University, Drs Bashaasha and Kiiza

Introduction

An assessment of regional and global banana markets was done within the framework of the Improvement of Banana Marketing and Utilization (IBMU) project. The first step was to put up a product differentiation according to the different varieties of bananas and banana products traded on the world market. It shows that the major part of bananas traded are dessert bananas (Cavendish or Gros Michel), with minor trade shares for other varieties such as cooking and roasting bananas (plantains). Although Uganda holds a large share of the global banana production, it only plays a minor role on the global banana market. Although a major producer of cooking banana, Uganda's major export product in the banana sector is dessert banana, of both the Gros Michel and the apple or baby banana type.

Methodology

Assessment was done mainly by using bi-variate descriptive analyses, as well as cost-benefit analysis for assessing comparative advantages of certain banana subsectors on export markets.

Findings

International banana markets will face major changes in the next few years, induced in particular by the banana market deregulation in the EU, and the growing demand in Central and Eastern Europe. While global banana prices are expected to declined due to the deregulation, and also due to the fact that bananas have a low income elasticity in high income market segments, some markets still seem to be prospering, like the above mentioned Eastern European market and the Middle East. Minor segments are those of organically produced bananas and fair trade bananas, with small market shares (2–5%) but high annual growth rates.

Other aspects of banana marketing in the Northern hemisphere are seasonal price fluctuations, whereby importers can obtain premia when trading during the Eastern and Christmas holiday seasons, and quality aspects, both consumer-borne ones, which differ from country to country, and regulation borne ones like phytosanitary issues. Due to presently poor quality management on the ground, i.e., in primary production, there have to be strong efforts to obtain and maintain steady export quality.

Based on the global market assessment, a comparative advantage analysis has been done for Ugandan producers targeting European markets. It shows that in conventional production of dessert bananas, there is no comparative advantage for Uganda, due to high production costs, high gathering and handling costs prior to exporting, and especially high

transport costs to Europe. On the other hand, there seems to be a comparative advantage for specialities like organic bananas or bananas flown out by air.

Concerning regional markets, there seems to be a potential for exporting bananas of both dessert and cooking bananas to Kenya, yet a lesser potential for exporting to Rwanda.

Agricultural policy priorities for improving rural livelihoods in Southern Africa

By Muchopa, C. (IITA/FANRPAN), V.M. Manyong, and L.M. Sibanda (FANRPAN)

Introduction

This paper synthesizes the findings from the policy module of the USAID/RCSA Improving Rural Livelihoods Project. This policy module was implemented by IITA and FANRPAN in four pilot SADC countries namely, Malawi, Mozambique, Tanzania, and Zambia. The major objective was to provide a critical policy analysis and dialogue leading to harmonization of selected policies within SADC countries. The studies are expected to provide the analytical basis for the design of agricultural policy strategies that contribute to unlocking constraints to commercialization and investment in the agricultural sector for a sustained economic growth.

Methodology

The study was based on secondary data. Additional primary data were collected in both quantitative and qualitative parameters. The analysis of data combined several techniques such as descriptive statistics, multivariate models, and domestic resource cost ratio analysis (DRC). Spatial analyses were applied to define development and crop domains.

Findings

Five development domains were defined. The spatial coverage of these domains indicates that limited pockets of southern Africa have very high potentials for agriculture under rainfed conditions. The results from the country studies show that agriculture remains the predominant source of livelihood and the engine for growth. The major problems constraining the performance of agriculture were highlighted as follows: poor access to inputs, poor communication, land degradation, over-dependence on rainfed agriculture, underdeveloped marketing system, high prevalence of HIV/AIDS and weak legislation, and lack of enforcement of laws/acts among others. Though attractive to domestic private investment, the agricultural enterprises in all countries were generally found to be less attractive to foreign private investment. High taxes and tariffs, poor infrastructure, bureaucracies, lack of policy incentives and adverse business environment were identified as inhibiting the performance of agribusiness enterprises. The major constraints to trade identified in the study relate to trade development, facilitation and promotion, and the poor coordination among institutions. Three areas were identified for policy intervention as follows: diversification of agricultural commodities, trade in domestic markets, and trade at the SADC level. The priority staple commodities that were identified in order of priority in all four countries taken together are maize, cassava, and rice. Cotton was recommended as the major industrial priority commodity common to all countries. Under rainfed conditions, few pockets of Southern Africa were found with high potentials for some of the main priority commodities. Five cross cutting components were identified to promote the diversification of SADC agricultural commodities and trade, namely research and technology transfer, human resources and institutional capacity building, market and export development,

improvements in delivery systems, market information and communications, as well as the coordination and harmonization of policies to reduce bureaucratic delays.

Promotion of targeted policy dialogues with stakeholders in research and policy agendas, farmers, and the agribusiness sector was recommended as ideal advocacy strategies alongside with policy briefs and personal follow-ups by well-placed advocates.

It is expected that the resulting strategies and priority areas for intervention will catalyze support for the flow of investment into the agricultural sector by Governments, the private sector, and other donors thereby pointing the direction to strategic decisions for a sustainable agriculture and diversified economic growth.

This synthesis paper and other outputs produced by the IITA/FANRPAN collaborative studies are intended to initiate dialogue amongst policy makers and major stakeholders in agricultural policy formulation within the study countries and in the SADC region as a whole.

Output 3. Methodologies for scaling up, and scaling out of IITA research innovations

Geospatial laboratory

By C. Legg

Spatial Databases for sub-Saharan Africa

Work continued through the year in accumulating spatial data from public domain and other sources in order to build up IITA spatial databases to support work throughout sub-Saharan Africa.

The 90-metre SRTM DEM for all of Africa became available early in the year, and the entire dataset was downloaded and processing started for individual countries.

Landsat images from the 2000 archive were downloaded as required for individual projects, and high-capacity online storage was purchased to accommodate the entire archive. Negotiations were completed with USGS and UNEP to acquire the complete 1990 and 2000 archives by copying from UNEP early in 2005

Recent administrative boundaries for Africa were obtained from FEWS, and compilation of a new continental boundary map commenced. Detailed administrative boundaries, roads, rivers, and soils were obtained from government agencies in Mozambique, and negotiations for similar data started in Zambia.

The latest gridded population estimates for Africa were obtained from CIESIN, and detailed data for the 2002 census obtained in Mozambique.

Mapping livelihoods and nutrition in Nigeria

Initial extrapolation of livelihood and child nutrition data was completed by April, and an interim report prepared for submission to a special session on poverty mapping at the ESRI user conference in San Diego in June. Re-processing of the data using Small Area Estimation was completed by September, and a final report submitted to FAO and World Bank.

Niger Delta Development Commission Master Plan

GIS input to the report on the IITA agricultural consultancy for the NDDC Master Plan was completed early in the year with regression of livelihood and food production data, and mapping of rural enterprises. Further processing of some data was undertaken late in the year as a demonstration for SAKSS.

USAID Southern Africa Livelihoods Program

A series of maps and PowerPoint presentations showing market-orientated development domains for IITA mandate crops were prepared for this program. GPS training in support of the Baseline Survey for this program was undertaken in Zambia, Malawi, Tanzania, and Mozambique. At the request of INIA staff in Mozambique, additional high-performance GPS units linked to PDAs were configured with software and map and image data, and supplied to INIA staff in Maputo.

Sub-Saharan Africa Challenge Program (SSA-CP)

From January to June the geospatial laboratory supported the final preparatory phase for this program, serving with ILRI in the Spatial Analysis Working Group in support of pilot learning site selection by ASARECA, SADC, and CORAF, and finally at a meeting with FARA.

Strategic Analysis and Knowledge Support System (SAKSS)

Following a series of e-discussions through the year, a meeting with USAID, IFPRI, and ILRI in Kampala in October resulted in agreement for USAID funding for a West Africa SAKSS node, hosted at IITA. National consultants were immediately hired to undertake surveys of data availability in Nigeria and Mali, recruitment of a WA-SAKSS coordinator was initiated, and planning commenced for a WA-SAKSS stakeholder workshop in February 2005.

Cameroon landscape modeling

The Geospatial Manager continued to oversee this research program in Cameroon, and maintained regular contact with the national consultant undertaking this work in Yaounde. A workshop to summarize ASB program highlights in Cameroon was attended in March, and at another in May organized by WWF for planning the tri-national park between Cameroon, Gabon, and Congo, a presentation of the modeling process was given.

Agro-Climatology

Maintenance of routine climate records at Ibadan, Kano, Cotonou, and Onne stations continued through the year. IITA technicians visited Kano and Cotonou to train staff in basic maintenance and recording, and installed new equipment as required. Two simple third-generation automatic weather stations were installed on Ibadan campus during the year for evaluation purposes, in order to guide purchase of additional automatic stations for all IITA sites. Discussions were held with IITA staff in Kampala and Maputo concerning weather station requirements.

Biometrics

Biometricians in Ibadan, Cotonou, Kampala, and Yaounde, attached to the Geospatial Laboratory, continued to provide routine support in SAS to IITA staff and collaborators through the year.

Research Database Management System (RDBMS)

An additional technician was recruited early in the year, to assist particularly with intranet and internet developments. Considerable advance was made during the year with IITA mandate crops in ICIS, and cooperation of IITA scientists involved in crop breeding improved dramatically. Intranet access to the ICIS prototype was made available to all participating scientists. Discussions were held with Contracts and Grants Office on combining their PROMIS database for new and active projects with the PMS originally developed as part of the RDBMS for past projects, and it was agreed that we would work towards this integration.

Staff development

The Senior GIS technician attended a six-week training course at ESRI in the USA on ArcGIS9 and the ArcIMS. All biometricians attended a SAS African workshop in South Africa. The RDBMS manager attended the annual workshop for ICIS users.

Training

SAS courses for IITA staff and collaborators were held in Yaounde and Ibadan during the year, and an external course was also organized at the Nigeria Root Crops Research Institute. A GIS course for IITA staff and associates was given in Kano in December.

Output 4. Research priority setting and impact assessment of innovations

Activities completed in 2004

Ex-ante evaluation of the potential economic, nutrition, and health benefits of biofortified cassava roots in West Africa.

By Manyong, V.M, A.S. Bamire (O.A.U.), M. Banea (PRONANUT, DRC), I.O. Sanusi (UI, Nigeria), G. Nahimana (PRONANUT, DRC), D.O. Awotide, A.G.O. Dixon, and I.O. Akinyele FBFI, Nigeria)

Introduction

Cassava is a major staple that supplies more than 50% of daily energy to more than 200 million persons in sub-Saharan Africa (SSA). Cassava roots are known to be low in micronutrients such as vitamin A, iron, and zinc. Micronutrient deficiencies threaten the lives of millions of poor households and those located in remote rural areas of SSA often not targeted by fortification programs. This paper presents results from an ex-ante evaluation of nutrition and health benefits of increased vitamin A status of cassava roots through biofortification for at risk target groups in West Africa (Nigeria) and Central Africa (Democratic Republic of Congo). The two countries are major producers and consumers of cassava.

Methodology

The study was based mainly on secondary data (demographic, subclinical prevalence, and other clinical symptoms of VAD, food consumption, food composition, research and development costs, and traits of cassava varieties). Descriptive statistics, the "Disabilities Adjusted Life Years (DALYs) approach, and cost-benefit analysis were used for data analysis.

Findings

The results for Nigeria revealed that more than 1.4 million years of healthy life are lost annually due to VAD amounting to a huge monetary loss of about US$1403.6 million. The conservatively estimated annual health gains associated with BC roots range between 98 770 and 295 539 years of healthy life for Nigeria. This translates into an internal rate of return as high as between 185.9% and 243.8%. When compared to the total value of DALYs lost due to VAD in the population, the total contribution of the biofortification process to the reduction of the health burden is about 7.0% in the pessimistic scenario and about 21.1% in the optimistic scenario. These estimates, which are based on average values, are irrespective of the additional benefits accruable to the technology developer. For DRC, the results show that more than 700 000 years of adjusted human healthy life are lost annually because of VAD in DRC. This corresponds to a loss of about US$ 700 million every year. This loss is higher than the budget allocated to the Ministry of Health in DRC in 2004. The biofortification of cassava roots would save between 4 and 37 thousand years of human life annually. The investment is profitable because it results in an internal rate of return of at least 142, which is above the interest rate of any private bank. The annual amount saved through the biofortification of cassava could then be assigned to other sectors to improve the social and economic conditions of the population in DRC. Such an investment would be able to contribute significantly to reducing expenditures on health in the country. This is very important in this post war period where the country significantly needs money for its reconstruction.

From the above results it could be concluded that a research and development effort aimed at the biofortification of cassava roots is a powerful strategy in the fight against hidden hunger from micronutrient deficiencies, which African governments at the national and local levels and international investors should support to improve the standard of living of the people in SSA.

Adoption and impact of improved cowpea varieties on rural livelihoods in Northern Nigeria

By Arega D. Alene, V.M. Manyong, A. Tipilda, and B.B. Singh

Introduction

The International Institute of Tropical Agriculture (IITA) has developed a number of improved dual-purpose cowpea varieties (e.g., IT90K-277-2, IT89KD-288, and IT93K-452-1) with generally high grain as well as fodder yields. Dissemination of these varieties started in 1993/94 and larger dissemination has been achieved through farmer-to-farmer seed diffusion, which has been underway since 1997. A survey was carried out during the 2003/04 cropping season to assess the adoption and impact of improved cowpea varieties on rural livelihoods in northern Nigeria.

Methodology

A total of 480 farmers from 24 villages growing improved cowpea in Kano and Kaduna States were randomly selected and surveyed during the 2003/04 cropping season. The sustainable rural livelihoods framework was adapted to reflect the role of agricultural technology, and based on the adapted framework, endogenous technology adoption and impact and social impact assessment analytical models were developed and used to examine adoption and impact of improved varieties.

Findings

The study has revealed that about 79% of the sample farmers in Kano and 61% in Kaduna adopted improved cowpea varieties. The adoption rate of the most preferred variety in Kano, IT90K-277-2, has grown from only 1.3% farmers in 1994 to 24% farmers in 1999 and 79% farmers in 2003. The adoption rate of the most preferred variety in Kaduna, IT93K-452-1, has grown from only 3% in 1999 to 20% in 2000 and 66% in 2003. Education of the head, access to credit, participation in extension, and farmers' perceptions of fodder yield and earliness of maturity had a positive and significant influence on the adoption and intensity of cultivation of improved cowpea varieties. The results further showed that farmers who perceived the attribute of earliness of maturity of the varieties had a 43% higher probability of adopting the varieties, relative to those who did not perceive this attribute. The results confirmed the critical role of improved cowpea varieties in consumption smoothing in northern Nigeria, owing to the early harvests at a time when farmers deplete their food stocks. Econometric analysis of the determinants of farm income revealed that adoption of the improved cowpea package as a whole, as opposed to adoption of only some components of the package, has a positive and significant impact on incomes. The results further showed that, on average, adoption of the package of improved cowpea raised incomes by US$305 per farmer per cropping season. Total cultivated land, livestock ownership, and the share of soybean also significantly influenced incomes. Results from a similar analysis of the determinants of food security showed that adoption of improved cowpea varieties had a positive and significant impact on food security; the adopters, relative to non-adopters,

have about 33% higher probability of being food secure. Social capital and the share of soybean also have a significant influence on food security. Farmers' own assessment revealed that additional income from adoption of improved cowpea varieties enabled adopters to acquire a range of productive and household assets and to make investments in education and healthcare of their children.

The impact of IITA-led biological control of major pests in sub-Saharan African agriculture: A synthesis of milestones and empirical results

By Arega D. Alene, P. Neuenschwander, V.M. Manyong, O. Coulibaly, and R. Hanna

Introduction

This work reviews and presents a synthesis of milestones and impact results of the IITA-led biological control of major pests in SSA agriculture. Since the 1970s, the International Institute of Tropical Agriculture (IITA), in collaboration with national programs in sub-Saharan Africa (SSA) and other international organizations, has been working on the biological control of several exotic pests, which have threatened the production of major commodities and the livelihoods of millions of people. IITA has been able to control devastating pests, which have inflicted damage on the major staple food crops of the poor, and has used innovative approaches to scaling-up biological control technology in SSA. It has enabled many SSA countries to establish strong national biological control programs.

Methodology

A number of empirical studies relating to the biological control of cassava mealybug, cassava green mite, mango mealybug, and water hyacinth have been reviewed and the results synthesized.

Findings

Economic impact studies revealed high benefits of biological control, indicating the considerable success of the programs. The estimated benefit-cost ratios for biological control are 200 : 1 to 740 : 1 for the cassava mealybug, 145 : 1 for mango mealybug, and 124 : 1 for water hyacinth. Net present values (or benefits) of US$1.7 billion for Nigeria, US$383 million for Ghana, and US$74 million for Bénin were computed for the biological control of the cassava green mite. These studies captured only the financial benefits of classical biological control. The overall impact is expected to be even higher, including the benefits to ecological and human health. The above benefits were achieved using innovative approaches to scaling-up environmentally friendly technologies through a wide partnership of institutions across continents and by building up the capacity of national programs. The results thus show that classical biological control is a cost-effective and sustainable option to lower economic and environmental losses due to pests. Countries in SSA should thus value and further strengthen their knowledge and capacity for biological control.

Improved cowpea package: intra-household impact

By A. Tipilda, A. D. Alene, V.M. Manyong, and B.B. Singh

Introduction

Cowpea is a popular crop in West Africa. It is a source of cash when sold, either raw or processed. As women do the processing, it plays a critical role in their empowerment. Cowpea fodder is a much higher protein source for animal feed than cereals, is easily stored and

sold during dry season for additional cash. Healthy livestock, in turn, produce more milk, meat, traction, and manure for the sustainability of crop production. However, despite the multiple roles cowpea plays in the farming system and livelihoods, the productivity level in farmers' fields remains low. To address some of these problems, IITA has developed a package of technologies that include improved dual-purpose (grain and fodder) high yielding varieties, new crop spacing arrangements, and a minimum use of inputs.

Methodology

The impact assessment data was collected between December 2003 and March 2004, ten years after the introduction of the technologies to farmers. A total of 24 villages were chosen out of those that had been exposed to the new technology (16 in Kano and 8 in Kaduna) and 20 households were randomly selected from each sampled village. Focus group discussions were carried out for men and women separately. Anthropometric measurements were taken of 703 children aged 12 months to 7 years to assess the nutritional impact. The bargaining household model was used as a guiding theoretical framework of this study. The model is useful in identifying the different preferences of individuals, but also takes into account the processes which both mitigate conflicts in such differences and processes which promote disparities within the household. The sustainable livelihoods framework was also used in understanding how different assets accrued as a result of the adoption of the improved cowpea package, have an impact to different individuals within the household.

Findings

Many farmers (68%) adopted the new technology. Higher yields were found among adopters (1200 kg/ha versus 500 kg/ha for non-adopters) and incomes increased by 134%. Additional income was obtained in two ways; from the sale of grain by the household head, and or women processing additional grain into bean cakes for sale. Income held by women was noted as contributing significantly to education (78% of wives of adopters), health (67%) and nutrition (65%) within the household. This also enabled the women to start up other additional or new petty trading businesses such as start up capital for sewing woven caps, leather products, as well as the sale of salt, and other goods in the village. This built up social networks to which they could revert in times of hardship within the family. Saratu Maalim from Bichi village sums this up: *"We are now able to have a little money which we contribute to a common pool among the women. This money is then used either for community activities or is handed over to member per member per month. You know our culture only allows women to be within the village and we are now working together."*

During the focus group discussions, the women considered indicators such as happy, active, and plump children as the most striking features in improved health care from the adoption of improved cowpea. The household benefited from the nutritional values of cowpea and also from the purchase of different types of food and better food availability throughout the year. During the group meetings, Rahima of Gawalde village reported: *"...previously, we had to eat pap (porridge). Now we can afford rice and occasionally meat and milk. Now during the dry period (crisis period) we may go without fish or meat but not without some rice or vegetables".*

These perceptions were confirmed by the results on anthropometric indices for children (Z scores of weight-for-age or WAZ, height-for-age or HAZ, and weight-for-height or WHZ). The results were calculated under three groupings of households: two years of adoption of improved cowpea (ADOP2), three years of adoption (ADOP3), and four years of adoption

(ADOP4). Overall, the average Z scores are within the ranges of a good nutritional status. No significant differences were observed between children of adopters and those of non-adopters for all the Z scores for ADOP2. One significant difference (WHZ) was observed for ADOP3 and two significant differences were observed (WAZ and WHZ) for ADOP4. The implication of these results is that children had a good nutritional status. This is consistent with perceptions by both wives of adopters and non-adopters. These results also show that as sustainable adoption occurs over time, differences are being observed between children of adopters and those of non-adopters.

Accrued benefits filtered through the community as the larger family could be assisted in times of hardship. Also, increased participation in social events raised the family status. This was seen as an important asset. Status has a great deal of meaning in terms of influencing collective decision-making.

For the women in this study their status in terms of their respectability, both within the household and community, was seen as the most important impact in terms of gender relations, and indeed countered other aspects such as education, mobility, and choice of occupation. They felt that there was less conflict in the household over limited resources and how these were allocated. Binta Shaibu from Santa Rago village sums it up: *"Now my husband and I can even sit down and decide what to do together, what to contribute to a community activity. In the past he controlled all the resources because he was not confident that I could make a decision on so little money."*

Ongoing activities

Impact analysis of adoption of improved soybean varieties and utilization in northern Nigeria

By Ojiako, I.O. (PhD Student, U.I), V.M. Manyong, and A. Ikpi (UI)

Introduction

Soybean research intensified in Nigeria following increased costs of animal protein and realization of potential plant sources to meet nutrient needs of low-income population. Low yield, pod shattering, and susceptibility to pests and diseases were among the agronomic deficiencies of earlier cultivars. New varieties mainly from the International Institute of Tropical Agriculture (IITA) and Soybean Utilization Project were introduced in northern Nigeria to stimulate agricultural production, raise incomes, promote food and nutrition security, and reduce poverty among rural dwellers. This study sought to determine households' adoption status of the innovations, identify factors that fostered and constrained adoption, and assess their impacts on welfare of households and communities. It also examined gender roles in production and utilization.

Methodology

Using multistage sampling techniques, 307 households were randomly drawn and interviewed from five villages. Data were collected using structured questionnaires and a number of participatory rural appraisal tools. Anthropometric measurements were made of children drawn from sampled households. Data were analyzed using descriptive statistics, test statistics, regression, and correlation analyses.

Findings

Results of descriptive statistics revealed that 60.9% of households adopted; 47.9% were in active farming age, 21–40 years; average age was 43.6 years and soybean farming experience, 8.6 years. Correlation analyses showed significant relationships between adoption status and households' soybean farming experience ($r = 0.19$, $p < 0.01$), level of education ($r = 0.18$, $p < 0.01$), and wealth status ($r = 0.12$, $p < 0.05$). From Tobit analysis, yield, expenditure on hired labor, distance to grains market, transport cost to major city, membership of associations, and exposure to extension services significantly ($p < 0.01$) influenced households' decisions on adoption and use intensities of improved varieties. Effects of a one-percent marginal increases in variables on elasticities of adoption and use intensities are yield, 0.5% and 0.6%, expenditure on labor, 0.4% and 0.5%, membership of associations, 0.1% and 0.2%, and exposure to extension services, 0.2% and 0.3%. Discriminate analysis revealed that social organization membership, exposure to extension, and expenditure on hired labor sufficiently separated adopters from non-adopters, accounting for 94.1% correct classification performance. On impact analysis, improved soybean significantly influenced households' income through effects on increased yield ($p < 0.05$), marketed surpluses ($p < 0.01$), higher prices ($p < 0.05$), and marketability. Higher incomes provided more durable assets, quality foods, and access to human capital development. A total of 85.3% women consumed at least one soybean-incorporated food per week. *Awara*, 93.9%, *gauda*, 75.9%, and *dawadawa*, 22.0%, were most-adopted soybean-incorporated foods. In general, soybean assisted in enhancing the natural resource base, food and nutrition, employment, assets, wealth, social status, and human capital development, and promoting women's awareness levels and socioeconomic empowerment.

Networks promoting person-to-person interactions were recommended for disseminating innovations. Enabling environment would encourage private sector participation in development of soybean markets while encouraging the use of hired labor for large-scale production and commercialization. New technologies should be spread alongside livelihoods-related investment options, for effective use of financial gains. Soybean remained valuable to every household and improved cultivars had been adopted for their superior agronomic and market qualities.

Planned activities in 2005 include the finalization of the PhD thesis and the defence.

Ex-ante evaluation of possible applications of biotechnologies at IITA

By Vercauteren K. (MSc Student, K.U. Leuven), E. Tollens, and Matty Demont (K.U. Leuven); V.M. Manyong, and I. Ingelbrecht (IITA)

Introduction

IITA conducts research on its mandate crops (bananas, plantains, cassava, maize, cowpea, soybean, yam), on farming systems and on natural resource management of the major agro-ecologies of sub-Sahara Africa. IITA also conducts a (modest) biotechnology research program. The objective is to help IITA in priority setting regarding its research program, and particularly regarding its investments in biotechnology research. The last EPMR recommended that IITA strengthen its priority setting in research.

Methodology

The study will apply a Benefit-Cost ratio, Net Present Value, Internal Rate of Return, Net Benefit-Investment ratio of project analysis and a proper discount rate, say 5% at constant

prices to measure the potential returns of investing into biotechnological programs. Data will come from secondary sources. Three crops are targeted for this study to cover a range of food and cash crops, annual and perennial crops, energy-rich and protein-rich crops as follows:

- Cowpea: Development of a transgenic crop with Bt-genes for resistance against *Maruca vitrata*
- Cocoa: Marker-assisted selection to develop new varieties with improved resistance/ tolerance against *Phytophthora, Mirid* and/or Swollen Soot
- Cassava: Development of transgenic cassava with resistance to Cassava Brown Streak Virus Disease.

The research began in late 2004. Milestones for 2005 include the finalization of data collection, data analysis, and writing of the MSc. thesis.

Impact assessment of new cassava varieties in Western Kenya

By Steffen Abele

Collaborators: Pheneas Ntawuruhanga, EARRNET, Hannington Obiero, KARI, Martin Odendo, KARI

Introduction

In 1995, the cassava mosaic virus caused a 60% drop in Western Kenya's cassava production with an estimated loss of US$ 10 million per annum. In a consequent effort to re-establish productivity of the cassava systems, several improved cassava varieties were released from 1997 to 2003 by various research stakeholders under the auspices of EARRNET. So far, about 16 new varieties have been distributed to farmers. Through this, one third of the production losses in the region could be recovered. Improved cassava varieties reach a yield level of 30 t/h, compared to 6 t/h obtained from traditional varieties.

Methodology

The impact study tries to assess the following questions: Above all is the question of what benefit the improved varieties have brought to both producers and consumers in the region. First assessments showed that 60% of the farmers had heard of the new varieties, but only ten percent had adopted them so far. This also raised the question of distribution of the benefits across farmers' social and economic strata. The next question is on the impact on cassava markets obtained by the productivity increase. Declining prices caused by a rapid increase of production, for example, could reduce the incentives for the adoption of the new varieties. Another market related question is how increased cassava productivity in Western Kenya spills over to its neighboring regions, central and eastern Kenya and eastern Uganda.

Another question is which of the above mentioned varieties were adopted and why. All of these varieties have different quality characteristics concerning their suitability for food consumption and industrial or feed use.

Findings

A preliminary assessment of the primary production systems shows that adoption of new varieties is higher than expected, they account for 30% of the varieties planted. However,

one single improved variety accounts for 25% of the varieties planted. There also seems to be a regional concentration of improved varieties, especially in those regions adjacent to the research station that developed and distributed the varieties. Improved varieties show about twice the yields than the traditional ones, yet yields per hectare are lower than expected, with about 4.5 tons for the traditional and 7.5 tons for the improved ones. This could be due to lower planting material inputs than expected, and also to a certain restriction of availability of improved planting material.

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